NEPHROLOGY TEACHING SERIES

Contrast-Associated AKI

Modern risk, modern prevention

Don't let fear of contrast delay a life-saving scan. Real risk is lower than you were taught.

Dr. Cheng • Premier Nephrology Medical Group

MS3/MS4 rotation teaching

Why we get consulted

Post-contrast Cr rise — is this contrast-induced acute kidney injury (CI-AKI), or coincidence?

CONSULT TRIGGERS

- Cr rise 24–48 h post-contrast
- Intra-arterial contrast (angio, PCI)
- eGFR < 30 needing CT or MRI
- Dialysis around contrast study
- Future contrast safety counseling
- Post-angio atheroemboli
- Gadolinium NSF concern

TEACHING PEARL

Most 'contrast-induced AKI' is coincidental. Don't skip a scan that changes management. Optimize volume when you can.

ACR Manual on Contrast Media (2025). UpToDate: Prevention of CI-AKI in adults (2026).

Definitions: post-contrast acute kidney injury (PC-AKI) vs CI-AKI

The ACR distinguishes correlation from causation.

POST-CONTRAST AKI (PC-AKI)

- General term: AKI developing within 48 h of contrast
- Descriptive — makes no claim about causation
- Includes CI-AKI and coincidental AKI
- Most clinical 'CI-AKI' is actually PC-AKI
- Common alternative causes: sepsis, hemodynamics, nephrotoxins, obstruction

CONTRAST-INDUCED AKI (CI-AKI)

- Causation attributed to contrast
- Typical pattern: Cr rise starts 24–48 h, peaks 3–5 d, recovers by day 7
- Definition: Cr ↑ ≥ 25% or ≥ 0.3 mg/dL within 48 h (KDIGO)
- Bland sediment usually (no casts, no cells)
- Non-oliguric in most cases
- **True CI-AKI rate in modern series: 1–6% (much lower than historical 10–30%)**

American College of Radiology (ACR) 2021 Manual on Contrast Media.

Risk factors

eGFR + intra-arterial + volume status drive risk.

Category	Highest risk	Lower risk
Kidney function	eGFR < 30 (highest); AKI currently	eGFR ≥ 45 (minimal risk for IV contrast)
Route	Intra-arterial (angio, PCI, transcatheter aortic valve replacement (TAVR))	Intravenous
Volume	Hypovolemia, hypotension, sepsis	Euvolemic, pre-hydrated
Comorbidity	Diabetes + CKD, CHF, multiple myeloma (debated)	Non-diabetic without CKD
Contrast agent	High-osmolar (obsolete)	Low-osmolar, iso-osmolar
Dose	Large volumes (>100 mL), serial studies within 24–72 h	Minimal necessary volume
Drugs	NSAIDs, aminoglycosides, amphotericin, diuretics (volume depletion)	Held for procedure

UpToDate: Prevention of contrast-associated AKI (2026). ACR Manual on Contrast Media (2025).

Prevention that works — and what doesn't

Volume status is everything. NAC and bicarb are not.

WORKS

- Isotonic IV hydration — NS or LR, 1–1.5 mL/kg/h for 6–12 h pre- & post-contrast
- Low-osmolar or iso-osmolar contrast (standard now)
- Minimize contrast volume
- Hold nephrotoxins around the procedure (NSAIDs, diuretics if volume allows)
- Optimize hemodynamics before the study
- Avoid repeat contrast within 24–72 h when possible
- **Intra-arterial: POSEIDON — left ventricular end-diastolic pressure (LVEDP)-guided hydration**

DOESN'T WORK (don't use)

N-acetylcysteine (NAC)

PRESERVE showed no benefit vs placebo — stop using.

Sodium bicarbonate

PRESERVE: no superiority over isotonic saline.

Prophylactic HD / hemofiltration

Removes contrast but has not shown kidney protection; do not use prophylactically.

Routine hydration for IV contrast + eGFR 30–59

AMACING: no benefit; consider omitting.

PRESERVE (NEJM 2018). AMACING (Lancet 2017). POSEIDON.

Atheroembolic disease — the real post-angio AKI

Different timeline, different clue set, worse prognosis.

CLUES

- AKI 1–4 weeks post-angiography (delayed onset, not 48 h)
- Livedo reticularis, blue toes, digital gangrene
- Eosinophilia (~80%), hypocomplementemia
- New visual changes (Hollenhorst plaques on fundus)
- GI ischemia, pancreatitis
- Often stepwise decline, not monophasic
- Biopsy: cholesterol clefts in arterioles

MANAGEMENT

- Supportive only — no proven disease-modifying therapy
- Statin (may stabilize plaque)
- Control BP, diabetes, lipids aggressively
- Wound care for skin lesions
- **Avoid further anticoagulation / angiography if possible**
- Prognosis: ~30% need dialysis; partial renal recovery possible
- Steroids: controversial, limited evidence

UpToDate: Atheroembolic disease of the kidneys (2026).

Gadolinium — a quick aside

NSF risk is minimal with modern agents. Don't refuse all MRI for low eGFR.

Agent group	Stability	NSF risk	Use in low eGFR
Group I (linear, non-ionic)	Unstable	Highest — classic NSF agents (gadodiamide, gadoversetamide)	**Avoid**
Group II (macrocyclic + stable linear)	Stable	Extremely low (few case reports)	Use if needed even at eGFR < 30
Group III (linear, ionic)	Intermediate	Low	Usually OK if eGFR ≥ 30; consider Group II first

Key points: • Group II gadolinium is considered safe at eGFR < 30 — risk of missed MRI diagnosis usually outweighs NSF risk. • Do not start or intensify dialysis solely for gadolinium; if already on HD, coordinate with the next practical session. • Retention in brain (basal ganglia, dentate) is mostly linked to linear agents — clinical significance remains unclear. • Avoid gadolinium in pregnancy when possible.

ACR Manual on Contrast Media (2025). ACR/NKF 2020 Consensus.

Landmark trials in contrast

These trials changed what we teach.

Trial	Year	Takeaway
PRESERVE	2018	NAC and bicarb both ineffective for CI-AKI prevention in high-risk (eGFR < 45) — definitive.
AMACING	2017	No pre-hydration vs NS hydration in eGFR 30−59 with IV contrast — NO difference.
POSEIDON	2014	LVEDP-guided hydration in cath lab reduced CI-AKI vs standard (intra-arterial context).
ACT	2011	NAC in angiography — no benefit. Confirmed in PRESERVE.
REMEDIAL II	2011	Ranolazine or RenalGuard — limited use in practice.
Davenport et al.	2013	Propensity-matched cohorts — IV contrast does NOT materially increase AKI at eGFR ≥ 30.

Full citations on references slide.

Common pitfalls

The errors that lead to refusing life-saving imaging or giving useless prophylaxis.

AVOID

- Refusing contrast at eGFR ≥ 30
- Ordering NAC or bicarb
- Calling every post-contrast Cr rise CI-AKI
- Missing atheroemboli
- Avoiding Group II gado at low eGFR
- Large NS boluses in volume overload
- Skipping the urine check

DO INSTEAD

- Weigh risk of NOT scanning
- Isotonic IV, minimal volume
- Check Cr 48 h; seek alternatives
- Post-angio: think atheroemboli
- Group II gado safe at low eGFR
- Euvolemic + eGFR > 45: often no prophylaxis
- Always UA + UACR

Clinical case

Think it through — answer is on the next slide.

VIGNETTE

A 70-year-old man with CKD (eGFR 28), DM, HTN is admitted with chest pain. Troponin positive; cardiology recommends urgent catheterization. Team calls nephrology to 'clear' him — worried about CI-AKI. BP 138/82, euvolemic by exam, UA bland.

QUESTION: What do you recommend regarding the cath? And what prophylaxis, if any?

Clinical case — answer

Key teaching points from the case.

ANSWER

Proceed with catheterization — the risk of NOT treating the ACS far outweighs CI-AKI risk. Pre-hydrate with isotonic IV (1 mL/kg/h × 6-12 h pre and post). Use low-osmolar contrast, minimize volume. Skip NAC and bicarbonate.

TEACHING

Modern CI-AKI rates are much lower than taught historically, especially for IV contrast. Here the indication is intra-arterial (higher-risk route) in eGFR < 30 (highest-risk kidney function) — so prophylaxis is warranted. POSEIDON supports LVEDP-guided hydration in the cath lab if available. PRESERVE definitively debunked NAC and bicarbonate — neither helps vs isotonic NS. If he develops a Cr rise post-procedure: 24-48 h onset, peak 3-5 d, recovery ~7 d supports CI-AKI vs alternatives. If Cr rises 1-4 weeks after cath, think atheroemboli (livedo, eosinophilia, cholesterol clefts on biopsy).

REFERENCES

Contrast-Associated AKI

Guidelines

ACR Manual on Contrast Media (2025)

ACR/NKF 2020 Consensus — Gadolinium in CKD/ESRD

KDIGO 2012 AKI Guideline (contrast section)

ESUR 2018 Guidelines on Contrast Media

Landmark Trials

PRESERVE — NEJM 2018

AMACING — Lancet 2017

POSEIDON — Lancet 2014

ACT — Circulation 2011

REMEDIAL II — Circulation 2011

Davenport et al. — Radiology 2013

UpToDate (2026 access):

Prevention of contrast-associated acute kidney injury • Pathogenesis, clinical features, and diagnosis of contrast-associated AKI • Gadolinium-based contrast agents and nephrogenic systemic fibrosis • Atheroembolic disease of the kidneys